

Problem Set 3

Series and Fourier Transforms

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Contents

1	Convergence of a Power Series	1
2	Fourier Coefficients	1
3	A Short Computation	1
4	Discussion Question	2
5	References	2

1 Convergence of a Power Series

Determine the radius of convergence R of the power series

$$\sum_{n=1}^{\infty} \frac{x^n}{n \cdot 2^n}.$$

Solution. Apply the ratio test to the coefficients $a_n = \frac{1}{n \cdot 2^n}$:

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{n \cdot 2^n}{(n+1) \cdot 2^{n+1}} = \lim_{n \rightarrow \infty} \frac{n}{2(n+1)} = \frac{1}{2}.$$

Hence $R = 1/(1/2) = 2$. The series converges absolutely for $|x| < 2$ and diverges for $|x| > 2$; the endpoints $x = \pm 2$ must be checked separately.

Endpoint check

At $x = 2$ the series becomes the harmonic series $\sum 1/n$ (divergent); at $x = -2$ it becomes the alternating harmonic series $\sum (-1)^n/n$ (convergent). So the interval of convergence is $[-2, 2)$.

2 Fourier Coefficients

Compute the Fourier sine-series coefficients b_n of $f(x) = x$ on the interval $(0, \pi)$, given

$$b_n = \frac{2}{\pi} \int_0^{\pi} x \sin(nx) \, dx.$$

Solution. Integrating by parts with $u = x$, $dv = \sin(nx) \, dx$:

$$\int_0^{\pi} x \sin(nx) \, dx = \left[-x \frac{\cos(nx)}{n} \right]_0^{\pi} + \frac{1}{n} \int_0^{\pi} \cos(nx) \, dx = \frac{-\pi \cos(n\pi)}{n} = \frac{\pi(-1)^{n+1}}{n}.$$

Therefore

$$b_n = \frac{2}{\pi} \cdot \frac{\pi(-1)^{n+1}}{n} = \frac{2(-1)^{n+1}}{n}.$$

3 A Short Computation

The partial sums above were checked numerically with the snippet below, which sums the sine series and compares it against $f(x) = x$ at a set of sample points.

```

1 import numpy as np
2
3 def sine_series(x, terms=50):
4     n = np.arange(1, terms + 1)
5     b = 2 * (-1) ** (n + 1) / n          # Fourier sine coefficients
6     return (b[:, None] * np.sin(np.outer(n, x))).sum(axis=0)
7
8 xs = np.linspace(0.1, np.pi - 0.1, 5)
9 err = np.max(np.abs(sine_series(xs) - xs))
10 print(f'max error over sample points: {err:.4f}')

```

The maximum error over the sample points falls off roughly like $O(1/\sqrt{N})$ in the number of terms N , consistent with the slow convergence expected near the jump discontinuity of the periodic extension.

4 Discussion Question

State it in your own words

Explain, in two or three sentences, why the Fourier series of $f(x) = x$ exhibits the Gibbs phenomenon near $x = \pi$, and what happens to the overshoot as the number of terms increases.

Answer. The periodic extension of $f(x) = x$ has a jump discontinuity at $x = \pi$ (from π down to $-\pi$). Near a jump, the partial sums overshoot the true value by a fixed fraction (9%) of the jump height no matter how many terms are taken – the overshoot narrows and moves toward the discontinuity as $N \rightarrow \infty$, but its height does not vanish.

5 References

- [1] Stein, E. M., & Shakarchi, R. *Fourier Analysis: An Introduction*. Princeton University Press.
- [2] Tolstov, G. P. *Fourier Series*. Dover Publications.